

Your grade: 100%

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Next item →

1. Which of the following equivalences is true?

1 / 1 point

- ☒ Sometimes it is easier to define an object in terms of the object itself
- ☐ It is always difficult to define a mathematical object explicitly
- ☐ The process of defining an object in terms of the object itself is called syllogism

Nice work

Correct. Recursion can simplify the definition of certain objects.

2. Which of the following statements is true about recursively defined functions?

1 / 1 point

- ☒ The basis step specifies an initial value of the function
- ☐ Defining a recursive function from N to R is not equivalent to defining a sequence
- ☐ The recursive step does not indicate how to find the value of the function at an integer from its values at smaller integers

Nice work

Correct. The basis step provides the starting value(s) for the recursive definition.

3. Find the recursive definition of the sequence a_n where $\forall n \in \mathbb{N} \ a_n = 5^n$.

1 / 1 point

- ☐ This sequence can be defined by setting $a_1 = 5$ and declaring that $\forall n \geq 1 \ a_{n+1} = 5 + a_n$
- ☐ This sequence can be defined by setting $a_1 = 5$ and declaring that $\forall n \geq 1 \ a_{n+1} = 5/a_n$
- ☒ This sequence can be defined by setting $a_1 = 5$ and declaring that $\forall n \geq 1 \ \forall n \geq 1 \ a_{n+1} = 5a_n$

Nice work

Correct. This correctly defines the sequence recursively.

4. Which of the following statements about recursively defined sets is true?

1 / 1 point

- ☐ It is always easier to define a set recursively than defining it explicitly
- ☐ The recursive step does not provide any rule for constructing new elements of the set from those we already have
- ☒ The basis step specifies some initial elements of the set

Nice work

Correct. The basis step provides the starting elements for the recursive definition.

5. Consider the subset S of the set of integers recursively defined by:

1 / 1 point

- basis step: $6 \in S$
- recursive: if $x \in S$ and $y \in S$, then $x + y \in S$

- ☐ S is the set of all positive integers that are multiples of 12
- ☐ S is the set of all positive integers divisible by 3
- ☒ S is the set of all positive integers that are multiples of 6

Nice work

Correct. The set defined includes all positive integers that are multiples of 6.